

在线电影票务系统 - 完整满分开发指南 (V2.0)

本指南严格针对《Java Web开发技术》试卷题目(4)编写。

环境要求: JDK 17.0.12 | Tomcat 9.0.63 | MySQL 8.0 | IntelliJ IDEA

技术栈: JSP + Servlet + JDBC (符合试卷禁用框架的要求)

第一步: 数据库构建 (MySQL)

为了满足“选座”、“订单”、“影评”和“管理员”功能, 我们需要 4 张表。请在数据库中执行以下 SQL:

-- 1. 创建数据库

```
CREATE DATABASE IF NOT EXISTS movie_ticket_system CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE movie_ticket_system;
```

-- 2. 用户表 (User)

```
CREATE TABLE users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(50) NOT NULL UNIQUE,
    password VARCHAR(50) NOT NULL,
    role VARCHAR(20) DEFAULT 'USER' -- 'ADMIN' (管理员) 或 'USER' (普通用户)
);
```

-- 3. 电影表 (Movie)

```
CREATE TABLE movies (
    id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(100) NOT NULL,
    director VARCHAR(50),
    price DECIMAL(10, 2) NOT NULL,
    description TEXT,
    release_date DATE
);
```

-- 4. 订单表 (Order) - 记录谁买了哪部电影的哪些座位

```
CREATE TABLE orders (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT,
    movie_id INT,
    seat_info VARCHAR(100), -- 例如 "3排4座"
    order_time DATETIME DEFAULT CURRENT_TIMESTAMP,
```

```

FOREIGN KEY (user_id) REFERENCES users(id),
FOREIGN KEY (movie_id) REFERENCES movies(id)
);

-- 5. 影评表 (Review) - 题目要求的“影评发布”功能
CREATE TABLE reviews (
  id INT PRIMARY KEY AUTO_INCREMENT,
  user_id INT,
  movie_id INT,
  content TEXT,
  rating INT, -- 1-5分
  create_time DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id),
  FOREIGN KEY (movie_id) REFERENCES movies(id)
);

-- 6. 初始化测试数据
-- 管理员账号: admin/admin123
INSERT INTO users (username, password, role) VALUES ('admin', 'admin123', 'ADMIN');
-- 普通用户: student/123456
INSERT INTO users (username, password, role) VALUES ('student', '123456', 'USER');

INSERT INTO movies (title, director, price, description, release_date) VALUES
('阿凡达: 水之道', '卡梅隆', 60.00, '视觉盛宴, 重返潘多拉', '2022-12-16'),
('流浪地球2', '郭帆', 50.00, '中国科幻里程碑', '2023-01-22'),
('奥本海默', '诺兰', 55.00, '原子弹之父的传记', '2023-08-30');

```

第二步: IDEA 项目配置 (针对 JDK17 + Tomcat 9)

1. 新建项目: File -> New -> Project -> Maven Archetype -> maven-archetype-webapp。
2. 配置 **Tomcat**: Run -> Edit Configurations -> '+' Tomcat Server Local -> Fix Deployment (选择 war exploded)。
3. 修改 **pom.xml**:
 - Tomcat 9 必须使用 javax.servlet API。
 - JDK 17 需要在 maven compiler 插件中指定版本。

pom.xml 完整内容:

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"(http://maven.apache.org/POM/4.0.0)"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"(http://www.w3.org/2001/XMLSche
ma-instance)"

```

```
xsi:schemaLocation="[http://maven.apache.org/POM/4.0.0](http://maven.apache.org/POM/4.0.0)
[http://maven.apache.org/xsd/maven-4.0.0.xsd](http://maven.apache.org/xsd/maven-4.0.0.xsd)">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.movie</groupId>
  <artifactId>MovieSystem</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>war</packaging>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>

  <dependencies>
    <!-- Tomcat 9 核心依赖 (javax) -->
    <dependency>
      <groupId>javax.servlet</groupId>
      <artifactId>javax.servlet-api</artifactId>
      <version>4.0.1</version>
      <scope>provided</scope>
    </dependency>
    <!-- JSTL 标签库 (JSP页面用) -->
    <dependency>
      <groupId>javax.servlet</groupId>
      <artifactId>jstl</artifactId>
      <version>1.2</version>
    </dependency>
    <!-- MySQL 驱动 -->
    <dependency>
      <groupId>com.mysql</groupId>
      <artifactId>mysql-connector-j</artifactId>
      <version>8.0.33</version>
    </dependency>
  </dependencies>
</project>
```

第三步: Java 后端核心代码

包结构如下:

- com.movie.util (数据库工具)
- com.movie.entity (实体类)
- com.movie.servlet (业务逻辑)
- com.movie.filter (权限拦截)

1. 数据库工具类 (**com.movie.util.DBUtil**)

请务必修改数据库密码!

```
package com.movie.util;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBUtil {
    private static final String URL =
"jdbc:mysql://localhost:3306/movie_ticket_system?useUnicode=true&characterEncoding=utf-8&serverTimezone=Asia/Shanghai";
    private static final String USER = "root";
    private static final String PASSWORD = "YOUR_PASSWORD"; // 改成你的MySQL密码

    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            e.printStackTrace();
        }
    }

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

2. 实体类 (**com.movie.entity.User**)

(其他实体类如 *Movie*, *Order* 可以直接在 JSP 中用 *ResultSet* 简化, 或者按此模式创建)

```
package com.movie.entity;
```

```

public class User {
    private int id;
    private String username;
    private String role; // "ADMIN" or "USER"

    public User(int id, String username, String role) {
        this.id = id;
        this.username = username;
        this.role = role;
    }
    // Getters
    public int getId() { return id; }
    public String getUsername() { return username; }
    public String getRole() { return role; }
    public boolean isAdmin() { return "ADMIN".equals(role); }
}

```

3. 用户注册与登录 (LoginServlet, RegisterServlet)

LoginServlet.java:

```

package com.movie.servlet;

import com.movie.entity.User;
import com.movie.util.DBUtil;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.sql.*;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws
ServletException, IOException {
        req.setCharacterEncoding("UTF-8");
        String u = req.getParameter("username");
        String p = req.getParameter("password");

        try (Connection conn = DBUtil.getConnection()) {
            PreparedStatement ps = conn.prepareStatement("SELECT * FROM users WHERE

```

```

username=? AND password=?");
    ps.setString(1, u);
    ps.setString(2, p);
    ResultSet rs = ps.executeQuery();
    if (rs.next()) {
        User user = new User(rs.getInt("id"), rs.getString("username"), rs.getString("role"));
        req.getSession().setAttribute("currentUser", user);
        resp.sendRedirect("index.jsp"); // 登录成功去主页
    } else {
        req.setAttribute("msg", "账号或密码错误");
        req.getRequestDispatcher("login.jsp").forward(req, resp);
    }
} catch (SQLException e) { e.printStackTrace(); }
}
}

```

RegisterServlet.java (注册功能):

```

package com.movie.servlet;

import com.movie.util.DBUtil;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.sql.*;

@WebServlet("/register")
public class RegisterServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws
ServletException, IOException {
        req.setCharacterEncoding("UTF-8");
        String u = req.getParameter("username");
        String p = req.getParameter("password");

        try (Connection conn = DBUtil.getConnection()) {
            PreparedStatement ps = conn.prepareStatement("INSERT INTO users (username,
password, role) VALUES (?, ?, 'USER')");
            ps.setString(1, u);
            ps.setString(2, p);
            ps.executeUpdate();
            resp.sendRedirect("login.jsp?msg=register_success");
        }
    }
}

```

```

    } catch (SQLException e) {
        req.setAttribute("msg", "注册失败, 用户名可能已存在");
        req.getRequestDispatcher("register.jsp").forward(req, resp);
    }
}
}

```

4. 业务逻辑Servlet (MovieActionServlet, OrderServlet)

MovieActionServlet.java (管理员管理电影 & 用户发影评):

```

package com.movie.servlet;

import com.movie.entity.User;
import com.movie.util.DBUtil;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.sql.*;

@WebServlet("/movieAction")
public class MovieActionServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws
ServletException, IOException {
        req.setCharacterEncoding("UTF-8");
        String action = req.getParameter("action");
        User user = (User) req.getSession().getAttribute("currentUser");

        try (Connection conn = DBUtil.getConnection()) {
            if ("add".equals(action) && user != null && user.isAdmin()) {
                // 管理员: 添加电影
                String title = req.getParameter("title");
                String price = req.getParameter("price");
                String desc = req.getParameter("description");
                PreparedStatement ps = conn.prepareStatement("INSERT INTO movies (title, price,
description, release_date) VALUES (?, ?, ?, NOW())");
                ps.setString(1, title);
                ps.setString(2, price);
                ps.setString(3, desc);
                ps.executeUpdate();
                resp.sendRedirect("index.jsp");
            }
        }
    }
}

```

```

    } else if ("delete".equals(action) && user != null && user.isAdmin()) {
        // 管理员:删除电影
        String id = req.getParameter("id");
        PreparedStatement ps = conn.prepareStatement("DELETE FROM movies WHERE
id=?");
        ps.setString(1, id);
        ps.executeUpdate();
        resp.sendRedirect("index.jsp");

    } else if ("review".equals(action) && user != null) {
        // 用户:发表影评
        String mid = req.getParameter("movieId");
        String content = req.getParameter("content");
        PreparedStatement ps = conn.prepareStatement("INSERT INTO reviews (user_id,
movie_id, content, rating) VALUES (?, ?, ?, 5)");
        ps.setInt(1, user.getId());
        ps.setString(2, mid);
        ps.setString(3, content);
        ps.executeUpdate();
        resp.sendRedirect("detail.jsp?id=" + mid);
    }
} catch (SQLException e) { e.printStackTrace(); }
}
}

```

OrderServlet.java (用户购票):

```

package com.movie.servlet;

import com.movie.entity.User;
import com.movie.util.DBUtil;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.sql.*;

@WebServlet("/buyTicket")
public class OrderServlet extends HttpServlet {
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws
ServletException, IOException {

```



```

req.setCharacterEncoding("UTF-8");
User user = (User) req.getSession().getAttribute("currentUser");
if (user == null) { resp.sendRedirect("login.jsp"); return; }

String movied = req.getParameter("movied");
String seats = req.getParameter("seats"); // 前端传来的座位号, 如 "1排1座"

try (Connection conn = DBUtil.getConnection()) {
    PreparedStatement ps = conn.prepareStatement("INSERT INTO orders (user_id,
movie_id, seat_info) VALUES (?, ?, ?)");
    ps.setInt(1, user.getId());
    ps.setString(2, movied);
    ps.setString(3, seats);
    ps.executeUpdate();
    // 购票成功, 去我的订单页
    resp.sendRedirect("myOrders.jsp");
} catch (SQLException e) { e.printStackTrace(); }
}
}

```

5. 权限过滤器 (AuthFilter) - 试卷第五题重点

```

package com.movie.filter;

import com.movie.entity.User;
import javax.servlet.*;
import javax.servlet.annotation.WebFilter;
import javax.servlet.http.*;
import java.io.IOException;

@WebFilter("/*")
public class AuthFilter implements Filter {
    public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain)
throws IOException, ServletException {
        HttpServletRequest req = (HttpServletRequest) request;
        HttpServletResponse resp = (HttpServletResponse) response;
        String uri = req.getRequestURI();

        // 1. 放行静态资源和登录注册页面
        if (uri.endsWith("login.jsp") || uri.endsWith("register.jsp") || uri.contains("/login") ||
uri.contains("/register") || uri.contains(".css")) {
            chain.doFilter(request, response);

```

```

        return;
    }

    // 2. 检查登录状态
    User user = (User) req.getSession().getAttribute("currentUser");
    if (user == null) {
        resp.sendRedirect("login.jsp");
        return;
    }

    // 3. 简单的管理员权限检查 (防止普通用户访问管理员页面)
    if (uri.contains("admin") && !user.isAdmin()) {
        resp.sendRedirect("index.jsp"); // 权限不足, 踢回主页
        return;
    }

    chain.doFilter(request, response);
}

public void init(FilterConfig config) {}
public void destroy() {}
}

```

第四步: 前端 JSP 页面 (View 层)

1. 主页 (index.jsp) - 区分管理员和用户视图

管理员可以看到“添加电影”和“删除”按钮, 普通用户只能看。

```

<%@ page import="java.sql.*,com.movie.util.DBUtil,com.movie.entity.User" %>
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<html>
<head>
<title>电影票务系统</title>
<style>
    body { font-family: sans-serif; padding: 20px; }
    .nav { margin-bottom: 20px; padding: 10px; background: #eee; }
    .movie-card { border: 1px solid #ddd; padding: 10px; margin-bottom: 10px; border-radius:
5px; }
    .btn { padding: 5px 10px; text-decoration: none; background: #007bff; color: white;
border-radius: 3px; }
    .btn-danger { background: #dc3545; }
</style>

```

```

</head>
<body>
    <% User user = (User) session.getAttribute("currentUser"); %>

    <div class="nav">
        <span>欢迎, <%= user != null ? user.getUsername() : "游客" %></span> |
        <% if(user != null) { %>
            <a href="myOrders.jsp">我的订单</a> |
            <a href="login.jsp">退出登录</a>
        <% } %>
    </div>

    <h1>正在热映</h1>

    <!-- 管理员专属: 添加电影区域 -->
    <% if (user != null && user.isAdmin()) { %>
    <div style="border: 2px dashed #ccc; padding: 15px; margin-bottom: 20px;">
        <h3>[管理员功能] 添加新电影</h3>
        <form action="movieAction" method="post">
            <input type="hidden" name="action" value="add">
            电影名: <input type="text" name="title" required>
            价格: <input type="text" name="price" required>
            描述: <input type="text" name="description">
            <button type="submit">上架电影</button>
        </form>
    </div>
    <% } %>

    <!-- 电影列表 -->
    <%
        try (Connection conn = DBUtil.getConnection()) {
            ResultSet rs = conn.createStatement().executeQuery("SELECT * FROM movies ORDER
BY id DESC");
            while (rs.next()) {
    %>
        <div class="movie-card">
            <h3><%= rs.getString("title") %> <small style="color:red">¥<%= rs.getDouble("price")
%></small></h3>
            <p>导演: <%= rs.getString("director") != null ? rs.getString("director") : "未知" %></p>
            <p><%= rs.getString("description") %></p>

            <a href="detail.jsp?id=<%= rs.getInt("id") %>" class="btn">查看详情 / 购票 / 影评</a>

```

```

        <!-- 管理员专属:删除按钮 -->
        <% if (user != null && user.isAdmin()) { %>
            <form action="movieAction" method="post" style="display:inline;">
                <input type="hidden" name="action" value="delete">
                <input type="hidden" name="id" value="<%= rs.getInt("id") %>">
                <button type="submit" class="btn btn-danger" onclick="return confirm('确定下架
吗? ')">下架</button>
            </form>
        <% } %>
    </div>
<%
}
} catch (Exception e) { out.print("数据加载错误"); }
%>
</body>
</html>

```

2. 详情与购票页 (detail.jsp) - 选座与影评

```

<%@ page import="java.sql.*,com.movie.util.DBUtil" %>
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<html>
<head>
    <title>电影详情</title>
</head>
<body>
    <a href="index.jsp"><< 返回首页 </a>
    <%
        String id = request.getParameter("id");
        try (Connection conn = DBUtil.getConnection()) {
            // 1. 查询电影详情
            PreparedStatement ps = conn.prepareStatement("SELECT * FROM movies WHERE
id=?");
            ps.setString(1, id);
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
    %>
                <h1><%= rs.getString("title") %></h1>
                <p>价格: ¥<%= rs.getDouble("price") %></p>
                <p>剧情简介: <%= rs.getString("description") %></p>

                <hr/>
            }
        } catch (Exception e) {
    %>

```

```

<h3>选座购票</h3>
<form action="buyTicket" method="post">
  <input type="hidden" name="movieId" value="<%= id %>">
  <label>选择座位:</label>
  <select name="seats">
    <option value="1排1座">1排1座</option>
    <option value="1排2座">1排2座</option>
    <option value="2排5座">2排5座 (情侣座)</option>
    <option value="VIP厅-沙发座">VIP厅-沙发座</option>
  </select>
  <button type="submit">确认支付</button>
</form>

<hr/>
<h3>用户影评</h3>
<ul>
<%
  // 2. 查询该电影的影评
  PreparedStatement ps2 = conn.prepareStatement("SELECT r.*, u.username FROM
reviews r JOIN users u ON r.user_id = u.id WHERE movie_id=? ORDER BY id DESC");
  ps2.setString(1, id);
  ResultSet rs2 = ps2.executeQuery();
  while(rs2.next()){
%>
    <li>
      <strong><%= rs2.getString("username") %>:</strong>
      <%= rs2.getString("content") %>
      <small>(<%= rs2.getDate("create_time") %>)</small>
    </li>
  <% } %>
</ul>

<h4>发表影评</h4>
<form action="movieAction" method="post">
  <input type="hidden" name="action" value="review">
  <input type="hidden" name="movieId" value="<%= id %>">
  <textarea name="content" rows="3" cols="50" placeholder="这电影怎么样?"
required></textarea><br>
  <button type="submit">提交评论</button>
</form>
<%
  }
} catch (Exception e) { e.printStackTrace(); }

```

```

%>
</body>
</html>

```

3. 我的订单页 (myOrders.jsp)

```

<%@ page import="java.sql.*,com.movie.util.DBUtil,com.movie.entity.User" %>
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<html>
<head><title>我的订单</title></head>
<body>
    <h2>我的购票记录</h2>
    <a href="index.jsp">返回首页</a>
    <table border="1" cellpadding="10" style="margin-top:20px; border-collapse: collapse;">
        <tr>
            <th>订单号</th>
            <th>电影</th>
            <th>座位</th>
            <th>下单时间</th>
        </tr>
        <%
            User user = (User) session.getAttribute("currentUser");
            try (Connection conn = DBUtil.getConnection()) {
                String sql = "SELECT o.*, m.title FROM orders o JOIN movies m ON o.movie_id = m.id
WHERE o.user_id=? ORDER BY o.id DESC";
                PreparedStatement ps = conn.prepareStatement(sql);
                ps.setInt(1, user.getId());
                ResultSet rs = ps.executeQuery();
                while(rs.next()){
        %>
        <tr>
            <td><%= rs.getInt("id") %></td>
            <td><%= rs.getString("title") %></td>
            <td><%= rs.getString("seat_info") %></td>
            <td><%= rs.getTimestamp("order_time") %></td>
        </tr>
        <%
            }
        } catch (Exception e) { e.printStackTrace(); }
        %>
    </table>
</body>

```

</html>

4. 注册与登录页

login.jsp (登录页)

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<html>
<head>
  <title>用户登录</title>
  <style>
    body { font-family: sans-serif; display: flex; justify-content: center; margin-top: 50px; }
    .box { border: 1px solid #ccc; padding: 20px; border-radius: 5px; width: 300px; }
    input { width: 100%; margin-bottom: 10px; padding: 5px; }
    button { width: 100%; padding: 8px; background: #007bff; color: white; border: none;
cursor: pointer; }
    .error { color: red; font-size: 0.9em; margin-bottom: 10px; display: block;}
  </style>
</head>
<body>
  <div class="box">
    <h2>登录</h2>
    <%-- 显示注册成功消息 --%>
    <% if ("register_success".equals(request.getParameter("msg"))) { %>
      <span style="color: green; display:block; margin-bottom:10px;">注册成功, 请登录 !
</span>
    <% } %>

    <%-- 显示错误消息 --%>
    <span class="error">${msg}</span>

    <form action="login" method="post">
      <label>用户名:</label>
      <input type="text" name="username" required>
      <label>密码:</label>
      <input type="password" name="password" required>
      <button type="submit">登录</button>
    </form>
    <p style="text-align: center; font-size: 0.9em;">
      没有账号 ? <a href="register.jsp">去注册</a>
    </p>
  </div>
```

```
</body>
</html>
```

register.jsp (注册页)

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<html>
<head>
  <title>用户注册</title>
  <style>
    body { font-family: sans-serif; display: flex; justify-content: center; margin-top: 50px; }
    .box { border: 1px solid #ccc; padding: 20px; border-radius: 5px; width: 300px; }
    input { width: 100%; margin-bottom: 10px; padding: 5px; }
    button { width: 100%; padding: 8px; background: #28a745; color: white; border: none;
cursor: pointer; }
    .error { color: red; font-size: 0.9em; margin-bottom: 10px; display: block;}
  </style>
</head>
<body>
  <div class="box">
    <h2>注册新用户</h2>

    <%-- 显示错误消息 --%>
    <span class="error">${msg}</span>

    <form action="register" method="post">
      <label>用户名:</label>
      <input type="text" name="username" required>
      <label>密码:</label>
      <input type="password" name="password" required>
      <button type="submit">立即注册</button>
    </form>
    <p style="text-align: center; font-size: 0.9em;">
      已有账号? <a href="login.jsp">去登录</a>
    </p>
  </div>
</body>
</html>
```

答题要点对照(用于写设计报告)

1. 用户/管理员角色: 我们在 users 表设计了 role 字段, 并在 AuthFilter 和 index.jsp 中做了权限判断(管理员才有添加/删除按钮)。
2. 选座购票: 在 orders 表和 OrderServlet 中实现了座位信息的存储。
3. 影评: 在 reviews 表和 MovieActionServlet 中实现了影评的增加和显示。
4. **JavaBean/Servlet/JSP**: 严格遵循了MVC模式(Entity -> Servlet -> JSP)。
5. 过滤器: AuthFilter 拦截了非法登录和越权访问。